

Ibrahim Muhammad

Backend Software Engineer | Node.js | NestJS | Python | DevOps | PostgreSQL

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Right to work: UK Student visa. Part-time employment permitted (up to 20 hours/week during term time; full-time during official university holidays).

PROFILE

Backend engineer with 4+ years' experience designing, building, and deploying production systems using Node.js, NestJS, TypeScript, and Python. Currently part of an early-stage UK travel-tech startup in ideation, focused on ancillary services that help passengers stay seated with family on flights. Experienced in multi-tenant SaaS architecture, CI/CD automation, and leading small engineering teams. MSc Data Science student at Heriot-Watt University, Edinburgh.

- Co-founding a seat-negotiation travel startup in Edinburgh, currently in ideation and product discovery
- Shipped 9+ production applications across education, HR, and public-sector clients
- Cut deployment time via GitHub Actions CI/CD and Docker
- Mentored 8+ junior developers and interns across 2 engineering teams

TECHNICAL SKILLS

Languages & frameworks: Node.js, NestJS, TypeScript, Python 3, Next.js, REST APIs	Databases & caching: PostgreSQL, Redis, Prisma ORM, multi-tenant data design
Messaging & queues: RabbitMQ, Redis Pub/Sub	Infrastructure & DevOps: Docker, Docker Compose, DigitalOcean, GitHub Actions, GitOps
Practices: Microservices, automated testing, code review, Agile delivery	AI & data: OpenAI API integration, LLM application design, data pipelines (MSc)

PROFESSIONAL EXPERIENCE

Backend Engineer (Part-time) <i>UK travel-tech startup, Edinburgh, Scotland (In-person)</i>	2026 – Present
• Part of a 4-person founding team developing an ancillary platform to help passengers negotiate airline seat assignments and stay seated with family or travel companions. • In ideation stage: defining the problem space, user journeys, and technical approach for seat negotiation workflows. • Shaping early backend architecture options, including API design and PostgreSQL data models for seat preferences and booking constraints.	
Backend Engineer & DevOps Engineer <i>Thermolinks Concept Ltd, Abuja, Nigeria (Remote)</i>	2023 – 2026
• Promoted from NYSC intern to full-time engineer within 12 months; built NestJS and TypeScript microservices powering the Nigeria Startup Portal and Project Noor. • Delivered a labelling and ticketing module that processed 2,000+ startup applications and cut manual status tracking for an 8-person operations team. • Moved email delivery onto RabbitMQ queues with retry logic, handling 500+ notifications daily and removing request timeouts caused by in-thread sends. • Paginated 25+ API endpoints and tuned PostgreSQL queries on datasets of 5,000+ rows, reducing average list response times by 60%.	

- Built an analytics module tracking application pipelines, ticket volumes, and team workload, giving leadership weekly visibility across 4 product areas.
- Automated CI/CD with GitHub Actions and Docker, reducing deployment time and eliminating manual release steps.

Backend Engineer (Contract)
Cynergy HR Solutions, Lagos, Nigeria (Remote)

2024 – 2025

- Developed core modules of a multi-tenant HR platform using Python, JavaScript, and NestJS, supporting 4 client organisations with isolated tenant data.
- Led the leave management module (requests, approvals, and audit trail), automating workflows for 1,200+ employees across client accounts.
- Designed multi-tenant onboarding that reduced new client setup from 2 weeks of manual configuration to 2 days via reusable tenant templates.

Web Manager (Voluntary, Part-time)
Sule Lamido University, Kafin Hausa, Nigeria (Remote)

2022 – Present

- Lead a team of 5 developers maintaining the university web estate and internal platforms using TypeScript, Next.js, NestJS, WordPress, and Laravel.
- Delivered an HR system covering 800+ staff records and a multi-tenant journal platform used across the entire university.
- Contributing to a new university digital framework planned to run the institution's web and internal systems for the next 5 years.
- Introduced code review standards and mentored 4 interns, improving pull request turnaround from 5 days to 2 days.

HIGHLIGHTED PROJECTS

Airline Seat Negotiation Platform

UK startup, 2026 – Present

- Early-stage ancillary travel concept helping families and groups stay seated together when airlines assign them apart. Currently in ideation and validation.

Nigeria Startup Portal: Labelling and Ticketing System

Thermolinks Concept Ltd, 2023 – 2024

- Public-sector onboarding portal processing 2,000+ startup applications with automated workflow tracking and status management.

Multi-Tenant Journal Management System

Sule Lamido University, Kafin Hausa, 2023 – 2024

- Single platform serving the entire university with isolated tenant data for journal publishing and management.

Cynergy HR: Leave Management Module

Cynergy HR Solutions, 2024 – 2025

- End-to-end leave workflow for 1,200+ employees with role-based approvals, balance tracking, and multi-tenant support.

EDUCATION

MSc Data Science

Jan 2026 – Present

Heriot-Watt University, Edinburgh, UK. On track for Distinction.

BSc Computer Science

2018 – 2023

Sule Lamido University, Kafin Hausa, Nigeria. First Class Honours equivalent (GPA 4.92/5.00). Best Graduating Student.

CERTIFICATIONS

Fundamentals of NestJS, Board Infinity / Coursera

Developing Back-End Apps with Node.js and Express, IBM / Coursera

Database Structures and Management with MySQL, Meta / Coursera

Generative AI: Prompt Engineering Basics, IBM / Coursera

Introduction to Software Engineering, IBM / Coursera

Code portfolio available on request. References available on request.